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REPORT

Assemblage structure of fishes and invertebrates inhabiting the coral *Pocillopora grandis* in Hawai‘i

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Abstract Coral reefs are highly diverse ecosystems, and the relationship between reef structural complexity and the diversity of resident species has long been of interest. Corals in the genus *Pocillopora,* especially one of the largest species, *P. grandis,* are characterized by highly branching morphology that hosts many fishes and macroinvertebrates. I documented the general environment, surrounding reefscape, and colony-specific variables most correlated with resident assemblage structure on *P. grandis* on the island of O‘ahu, Hawai‘i. In general, surrounding reefscape had little effect on total assemblage structure, with most variation attributed to colony size, percent living tissue, and general environmental factors associated with each reef. Resident fish and invertebrate abundance, Shannon–Wiener composite α-diversity, species evenness, and species richness increased with colony size, and fish abundance and species richness increased with greater living colony tissue. However, invertebrate abundance and species richness remained unchanged across equally sized living and dead colonies, although live colonies supported more coral obligate decapods and dead colonies more facultative species. Live colonies detached from the seafloor yet otherwise mostly intact hosted more abundant and diverse residents than fully intact, attached colonies. The capability of *P. grandis* to grow to large sizes, host diverse resident assemblages, and tolerate moderate physical disturbance, after which assemblages may become larger and more diverse due to increases in habitat heterogeneity, makes this coral an ideal candidate for reef restoration efforts in Hawai‘i.

Keywords Community ecology · *Pocillopora grandis* · Hawai‘i · Habitat effects · Coral reef degradation · Species diversity

Introduction

Tropical coral reefs are characterized by the highest species richness of any marine ecosystem, despite covering less than 1% of the global seafloor (Fisher et al. 2015). This diversity is partly due to the complex physical structure provided by corals (Bellwood et al. 2017), which offers refuge for fishes (Holbrook et al. 2008a) and mobile invertebrates (Idjadi and Edmunds 2006), thus mediating the effects on resident species of competition (Bonin et al. 2009) and predation (Beukers and Jones 1998). At the reefscape scale, species richness of fishes increases with greater habitat heterogeneity (Wilson et al. 2007) and coral species richness (Komyakova et al. 2013). On an individual coral, resident assemblage structure is affected by colony size and morphology (Abele and Patton 1976), coral identity (Richardson et al. 2017), and surrounding habitat rugosity (Darling et al. 2017).

Coral reefs are also among the most endangered ecosystems in the world (Bellwood et al. 2004). The tight associations between coral colonies and their resident fishes and invertebrates are under threat from a myriad of global and local stressors, including ocean warming (Hughes et al. 2017), overfishing of important functional groups (MacNeil et al. 2015; Shantz et al. 2020), sedimentation (Fabricius

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2005), and eutrophication (Fabricius 2011). While coral reefs in Hawai‘i have not yet been as severely affected by global stressors as other regions (Sully et al. 2019), annual bleaching is projected by mid-century (van Hooidonk et al. 2016) and the reefs are heavily exposed to local stressors that scale with human population density (DeMartini et al. 2008). Reefs around the Main Hawaiian Islands (MHI) experience a gradient of human influence, with those around O‘ahu being the most degraded (Friedlander et al. 2018).

Without continental shelves, Hawaiian coral reefs are naturally and greatly influenced by wave exposure (Dollar 1982; Grigg 1983; Franklin et al. 2013). The southern and western facing shores of O‘ahu were particularly affected by two major hurricanes in 1982 and 1992 that vastly diminished coral cover at many sites, with limited recovery (Coles and Brown 2007; Friedlander et al. 2008). Pocilloporid corals are most commonly found in reef habitats along the west- and south-facing shorelines of O‘ahu and are important branching habitat for both small fishes (Shima et al. 2008) and mobile invertebrates (Counsell et al. 2018).

Most pocilloporid species in Hawai‘i do not exceed 30 cm in diameter. However, *Pocillopora grandis* can grow to approximately 1.5 m in diameter, making it the most prominent coral species providing high relief and habitat complexity on exposed O‘ahu reefs (pers. obs.). Prior studies elsewhere in the Pacific have investigated *P. grandis* with regards to damselfish recruitment (Schmitt and Holbrook 2000) and habitat use (Sakanoue and Sakai 2019), hawkfish habitat preference (Kane et al. 2009), and a multitude of mutualisms. Mutualist residents increase coral growth rates through fertilization by fish excreta (Holbrook et al. 2008a, 2011), defend against invertebrate corallivores (McKeon and Moore 2014), and bolster resilience to bleaching (Shantz et al. 2023). Coker et al. (2014) found that, despite not being a thicket-forming species, *P. grandis* is among the top 10% of studied corals in terms of resident fish diversity. However, no studies of *P. grandis* have been conducted in Hawai‘i, and the drivers of resident assemblage structure are poorly understood globally.

Given the threats facing coral reefs and ongoing degradation of Hawaiian reefs near large human populations, it is important to understand the factors that structure resident fish and macroinvertebrate assemblages on *P. grandis*. Such knowledge fosters an understanding of how these assemblages may change under future conditions and whether this species could be leveraged for coral restoration efforts. Therefore, I explored the general environmental, surrounding reefscape, and colony-specific characteristics that potentially drive resident assemblage structure on *P. grandis* colonies. Specific questions addressed were: (1) How does assemblage structure differ relative to variations in a) site-specific environments, b) surrounding reefscape and c) colony-scale characteristics? (2) Is there evidence that the presence of particular species affects assemblage structure? and (3) How does physical disturbance affect resident assemblage structure?

Materials and methods

Study sites

Surveys were conducted at three sites along the south and west facing shores of O‘ahu, Hawai‘i (Fig. 1). The Hanauma Bay site (21°16′7.35″N, 157°41′34.47″W; ~ 20 ha, 60% hardbottom, 40% sand, 68 surveyed colonies) ranged in depth from 5 to 18 m and is located on the southeast side of the island. A fully protected Marine Life Conservation District since 1967, this continuous reef had the highest abundance of large-bodied fishes (Friedlander et al. 2019), highest coral cover (Franklin et al. 2013), and greatest coral diversity that included *Porites* spp., *Pocillopora* spp., and *Montipora* spp. (pers. obs.). The Waikīkī site was located approximately 800 m offshore of Waikīkī Beach along the Halekulani Sand Channel (21°16′11.00″N, 157°50′18.53″W; ~ 10 ha, 60% hardbottom, 40% sand, 92 surveyed colonies). This spur-and-groove coral reef had a comparable depth to Hanauma Bay, was dominated by colonies of *Porites lobata* and *Pocillopora meandrina*, had the lowest coral cover of the three sites (Franklin et al. 2013), and lacked larger bodied fishes, likely due to fishing and chronic human influence (Friedlander et al. 2018). The Kahe Point site (21°21′13.37″N, 158° 7′54.83″W; ~ 1 ha, 90% hardbottom, 10% sand, 50 surveyed colonies) was slightly shallower than the others (2–14 m) with coral cover intermediate to Hanauma Bay and Waikīkī and dominated by the same coral species as Waikīkī.

Pocillopora grandis surveys

Visual surveys of *Pocillopora grandis* colonies (n = 502 surveys, 210 colonies) were conducted on SCUBA from 2017 to 2023 (Table S1). Surveys were conducted on colonies ranging from 0 to 100% live tissue and in a variety of colony conditions (Fig. S1): (1) fully intact and attached to the seafloor (n = 435 surveys, 183 colonies), (2) detached from the seafloor and leaning but otherwise intact (n = 39 surveys, 23 colonies), (3) detached and split into 2–4 large pieces (n = 14 surveys, 9 colonies), and (4) detached and highly fragmented (n = 5 surveys, 5 colonies). All consecutive surveys of the same colony occurred at least 50 days apart unless a major change in condition occurred.

Surveys included a full and concurrent visual census of all resident fishes and macroinvertebrates. While open-circuit SCUBA has been shown to affect fish behavior and abundance following a survey (Emslie et al. 2018), this primarily affects targeted species (Gray et al. 2016) that do not

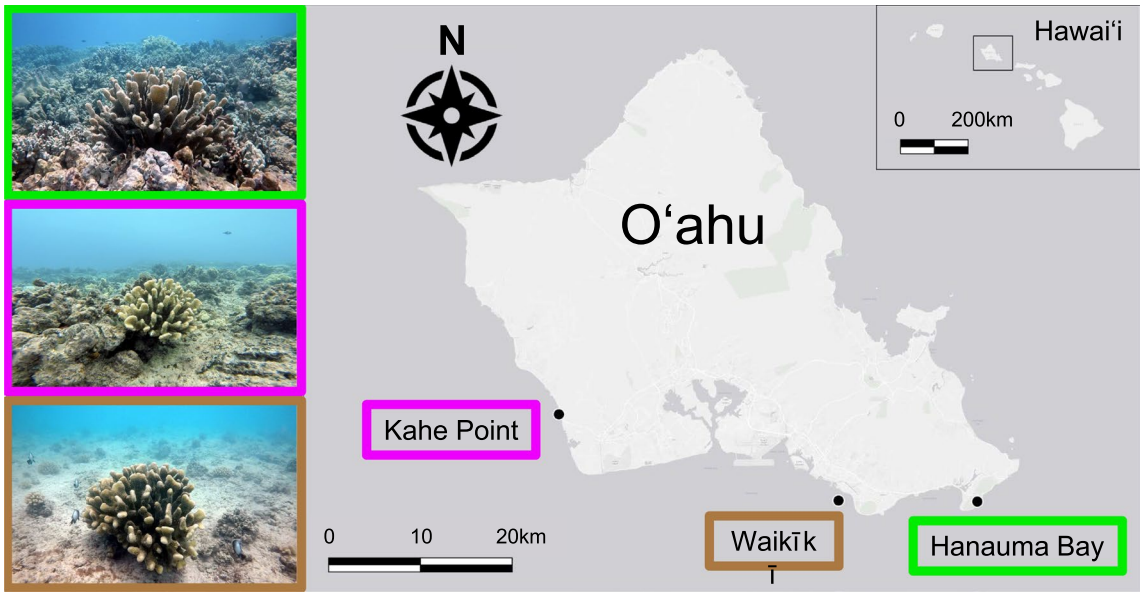

Fig. 1 Map of O'ahu, Hawai'i, with the locations of the three study sites. Photos are of *Pocillopora grandis* coral colonies with the typical surrounding reefscape at each site in order of increasing degradation and decreasing habitat complexity (Hanauma Bay: green, top; Kahe Point: magenta, middle; Waikīkī: brown, bottom)

regularly inhabit *P. grandis* (pers. obs.). To reduce disturbance, surveys were initiated 5 m away to record all non-resident individuals that interacted with but whose home range was greater than each singular colony. This was followed by a thorough search of all interstitial spaces with a dive light to record invertebrates and resident fishes that remained with the host coral for the duration of the survey, including cryptobenthic fishes. I could not regularly identify hermit crabs to species in situ, so these were grouped into a single higher taxonomic level. Live colony tissue was visually estimated in situ to the nearest 5%. Given the large range of colony sizes, I calculated the volume of an elliptical cylinder $\left(\text{volume} = \pi \times \left(\frac{\text{length}}{2}\right) \times \left(\frac{\text{width}}{2}\right) \times \text{height}\right)$ for a subset of surveys with measurements to the nearest 1 cm of maximum diameter (length), maximum orthogonal diameter (width), and height to estimate the amount of habitat space provided by each colony. I began measuring inter-branch spacing in 2018 (no measurements occurred at Kahe Point) by averaging the distance to the nearest 0.5 cm of five random branch tips to the nearest adjacent branch tip (see Counsell et al. 2018 for similar measurements of colony size and interstitial space). Finally, I recorded a video of the surrounding reefscape following each survey as a time-efficient approach to visually estimate the structural complexity and dominant benthos in an approximate 5 m radius around each colony. Occurrences of overlapping radii between *P. grandis* colonies was rare, and any fishes that occupied multiple colonies were not recorded as permanent residents. Variables describing structural complexity included the proportions estimated to the nearest 10% of the surrounding habitat that contained three-dimensional complexity from any source ("Complexity" in figures), *Pocillopora meandrina* ("POME"), mounding *Porites* spp. ("POLO"), *Porites compressa* ("POCO"), hard substrate without living coral ("Hard Non-coral", hereafter referred to as hardbottom in text), rubble, and sand. Additionally, I characterized the surrounding reefscape as shorter, approximately the same height, or taller than the focal coral ("Relief").

Statistical analysis

Reefscape variation

All statistics were completed within the software *R* (R Core Team 2023). To characterize the reefscape surrounding each colony, I ran a Principal Component Analysis (PCA) on the first survey of all intact colonies (n = 180). A combination of habitat characteristics that loaded most strongly on the first two PCA axes (i.e., "Complexity", "Hard Non-coral", and "Sand"; Fig. S2a) were used in all following univariate and multivariate analyses (see Table S2). I included *Pocillopora meandrina* coverage ("POME") because it occupied unique ordination space and is related and structurally similar to *P. grandis*. Because "Relief" was measured in tandem with "Complexity" to describe surrounding structure and isolation, this metric was included. Lastly, depth was included because it can affect species distributions (Counsell et al. 2018).

Assemblage structure on intact colonies

Mixed effects models were used to evaluate the effects of colony and reefscape characteristics on resident assemblage structure. Models included fixed effects for site, depth, colony volume, inter-branch spacing, living tissue, and reefscape characteristics that loaded most strongly or occupied unique ordination space (see above), as well as random effects for colony identity and survey date to account for repeated measures. Surveys were restricted to intact colonies still attached to the seafloor with all measured colony and reefscape characteristics (n = 263 surveys, 180 colonies). Response variables included resident fish and macroinvertebrate abundance, species richness (*S*), species evenness (*J'*, Pielou 1966), and Shannon–Wiener composite α-diversity (*H'*). Abundance and species richness were modeled with generalized linear mixed models (GLMMs) with negative binomial and Poisson error distributions, respectively, in the package *glmmTMB* (Brooks et al. 2017). Species evenness was logit-transformed $\left(\log\left(\frac{p}{1-p}\right)\right)$, where *p* is evenness on a scale of 0–1. Both evenness and diversity were modeled with linear mixed models (LMMs) with the *lme4* package (Bates et al. 2015). Colony volume was log-transformed to meet model assumptions, and all continuous predictor variables were scaled and centered. Predictor variable significance was determined by type III ANOVAs with Satterthwaite's method from the package *lmerTest* (Kuznetsova et al. 2017) for LMMs and χ^2 likelihood-ratio tests for GLMMs.

To evaluate the effects of reefscape and colony characteristics on assemblage structure within a multivariate framework, distance-based redundancy analysis (db-RDA, function *capscale*) (Legendre and Anderson 1999) and PERMANOVA (Bray–Curtis dissimilarity index, 999 permutations) were used, both in the *vegan* package (Oksanen et al. 2022). Most colonies (61.7%) were surveyed only once in this data set. Colonies with multiple surveys (Hanauma Bay: 39, Kahe Point: 0, Waikīkī: 30) had 575 ± 227 days (mean ± SD; range: 254–1537) between surveys, representing de facto statistically independent colonies that had grown substantially and hosted different assemblages. Because inter-branch spacing was marginally significant but measured at only two sites, analyses were conducted on two datasets: (1) surveys including these measurements (n = 137, 114 colonies, 2 sites) to test this variable's effect, and (2) all surveys (n = 258, 179 colonies, 3 sites) with this variable excluded. Assemblage data were square-root transformed and Wisconsin standardized, and predictor significance determined through permutation tests.

Species co-occurrence

Assemblage structure was further characterized with a species co-occurrence analysis (package *cooccur*; Griffith et al. 2016) based on the probabilistic model of Veech (2013) on the same subset of surveys as the full multivariate analyses. Due to the large variation in individual colony volume, surveys were binned into volume ranges. A scatter plot of all measured intact colony volumes revealed an inflection point that was designated as the largest size class ("Extra-Large" > 0.18m^3). All surveys of colonies less than this volume were equally divided into three smaller categories: "Small" (0.002–0.022m^3), "Medium" (0.022–0.05m^3), and "Large" (0.05–0.18m^3). I ran combined co-occurrence analyses on small to medium colonies and large to extra-large colonies.

Results

Assemblage characterization

Across all 502 surveys of 210 colonies at 3 sites, 10 fish and 14 invertebrate species were found in at least 5% of surveys (Table 1, see Table S3 for a complete list of species). Two families (Pomacentridae and Cirrhitidae) comprised 60% of the fish species, and nearly 80% of the invertebrates were decapods, including five species of guard crabs (*Trapezia* spp.). Nearly 80% of species occurred at all three sites, and all but one (the puffer *Canthigaster jactator*) occurred on the full range of colony sizes. In total, 3,558 individual fish, the vast majority of which were adults (92%), and 4,550 invertebrates were recorded. Of these, four fish species were present in at least 20% of surveys: two damselfishes (*Plectroglyphidodon johnstonianus* and *Dascyllus albisella*), an orbicular velvetfish (*Caracanthus typicus*), and a hawkfish (*Paracirrhites arcatus*). Eight invertebrate species were present in at least 20% of surveys, including all five species of trapezid guard crabs in Hawai'i (*Trapezia flavopunctata*, *T. intermedia*, *T. digitalis*, *T. bidentata*, *T. tigrina*), the murex snail *Galeropsis monodonta*, the coral shrimp *Harpiliopsis depressa*, and hermit crabs of the Superfamily Paguroidea.

Reefscape-level effects on assemblage structure of intact colonies

Principal Component Analysis revealed reefscape-scale differences among sites in the seafloor habitat surrounding surveyed corals (Fig. 2), with the first two component axes explaining 58% of the variance (Fig. S2a,b). Each site occupied a distinct position in ordination space. Hanauma Bay was most closely associated with greater total coral cover (sum of all species; mean ± SD; 26.8 ± 23.0%) and complexity (40.8 ± 25.9%), as well as less hardbottom (24.8 ± 19.6%) than the other sites. Waikīkī, at the other extreme, had greater cover of hardbottom (78.2 ± 23.4%), and lower live coral cover (8.7 ± 9.3%) and complexity (15.9 ± 12.4%) than

Table 1 Fishes and invertebrates seen in at least 5% of surveys of *Pocillopora grandis* colonies (n = 502 surveys of 210 colonies), listed in descending order of % occurrence

Family	Species	Species code	Sites present	% of surveys present	#/survey when present (μ±SE)	Coral vol. range (m^3)	Coral % live range	Coral physical state
Fishes								
Pomacentridae	*Plectroglyphidodon johnstonianus*	PLJO	H,K,W	59.8	1.9±0.07	S–XL	5-100	0,1,2,3
Cirrhitidae	*Paracirrhites arcatus*	PAAR	H,K,W	45.6	1.3±0.03	S–XL	0-100	0,1,2,3
Caracanthidae	*Caracanthus typicus*	CATY	H,K,W	45.4	2.1±0.08	S–XL	40-100	0,1,2,3
Pomacentridae	*Dascyllus albisella*	DAAL	H,K,W	23.9	7.9±0.91	S–XL	0-100	0,1,2,3
Tetraodontidae	*Canthigaster jactator*	CAJA	H,K,W	13.9	1.6±0.11	M–XL	0-100	0,1,2,3
Labridae	*Pseudocheilinus tetrataenia*	PSTE	H,K,W	9.2	1.3±0.09	S–XL	5-100	0,1,2,3
Cirrhitidae	*Amblycirrhitus bimacula*	AMBI	H,K,W	9.0	1.2±0.05	S–XL	45-100	0,1,2,3
Cirrhitidae	*Cirrhitops fasciatus*	CIFA	H,K,W	7.4	1.3±0.08	S–XL	0-100	0,1,2,3
Scorpaenidae	*Sebastapistes coniorta*	SECO	H,K,W	7.2	3.0±0.53	S–XL	50-100	0,1,2,3
Cirrhitidae	*Paracirrhites forsteri*	PAFO	K,W	7.0	1.2±0.07	S–XL	75-100	0,1,2
Invertebrates								
Trapeziidae	*Trapezia flavopunctata**	TRFL	H,K,W	64.9	2.8±0.14	S–XL	40-100	0,1,2,3
Trapeziidae	*Trapezia intermedia**	TRIN	H,K,W	48.6	1.7±0.04	S–XL	5-100	0,1,2,3
Muricidae	*Galeropsis monodonta*	GAMO	H,K,W	45.2	3.6±0.25	S–XL	40-100	0,1,2,3
Palaemonidae	*Harpiliopsis depressa*	HADE	H,K,W	38.6	3.4±0.32	S–XL	45-100	0,1,2,3
Trapeziidae	*Trapezia digitata**	TRDI	H,K,W	36.5	1.8±0.09	S–XL	40-100	0,1,2,3
Trapeziidae	*Trapezia bidentata**	TRBI	H,K,W	34.9	1.5±0.04	S–XL	50-100	0,1,2,3
Trapeziidae	*Trapezia tigrina**	TRTI	H,K,W	24.9	1.5±0.06	S–XL	60-100	0,1,3
Paguroidea **	unidentified hermit crabs	PAGU	H,K,W	23.7	2.8±0.40	S–XL	0-100	0,1,2,3
Ophiocomidae	*Breviturma pica*	BRPI	H,K,W	16.3	1.8±0.14	S–XL	0-100	0,1,2,3
Alpheidae	*Alpheus lottini**	ALLO	H,K,W	10.2	1.1±0.04	S–XL	40-100	0,1
Portunidae	*Goniosupradens hawaiensis*	GOHA	H,W	6.0	1.3±0.15	S–XL	50-100	0,1,2
Domeciidae	*Domecia hispida*	DOHI	H,K,W	5.6	1.7±0.20	S–XL	40-100	0,1,2,3
Stenopodidae	*Stenopus hispidus*	STHI	W	5.2	1.9±0.17	S–XL	0-100	0,1,2
Ophiocomidae	*Ophiocoma erinaceus*	OPER	H,K,W	5.0	1.6±0.23	S–XL	0-100	0,1,2,3

† superfamily
* known coral mutualist

Species codes are colored for multivariate ordinations (bold: fishes, italic: trapezid guard crabs, bolditalic: other invertebrates). Sites include Hanauma Bay (H), Kahe Point (K), and Waikīkī (W). Coral colony volume: S (small); M (medium); L (large); XL (extra-large; see Methods for size designations). Coral physical state: 0 (intact), 1 (intact but detached from seafloor and leaning on side), 2 (detached from seafloor and split into 2–4 leaning sections), 3 (detached from seafloor and highly fragmented). See Table S3 for all fishes and invertebrates found in at least 1% of surveys

the other sites. Kahe Point was intermediate in all measurements (coral cover: 15.1 ± 12.9%; complexity: 38.0 ± 20.2%; hardbottom: 56.6 ± 23.1%). All sites had similar coverage of *Pocillopora meandrina*.

Reefscape level factors had very few significant effects on univariate analyses of assemblage structure. Only relief had a significant negative effect on *P. grandis* resident fish abundance (GLMM: $\chi^2 = 6.69$, $p = 0.035$). However, site was a significant predictor for fish abundance (GLMM: $\chi^2 = 7.85$, $p = 0.020$) and invertebrate diversity (LMM: $F(2, 125.3) = 4.33$, $p = 0.015$), species richness (GLMM: $\chi^2 = 8.66$, $p = 0.013$), and abundance (Figs. 3, S3 left) (GLMM: $\chi^2 = 12.63$, $p = 0.002$; see Fig. S4 for Fig. 3 plotted on a linear scale; see Tables S4 and S5 for all univariate model fixed and random effect statistics). Colonies at Waikīkī had greater fish abundance

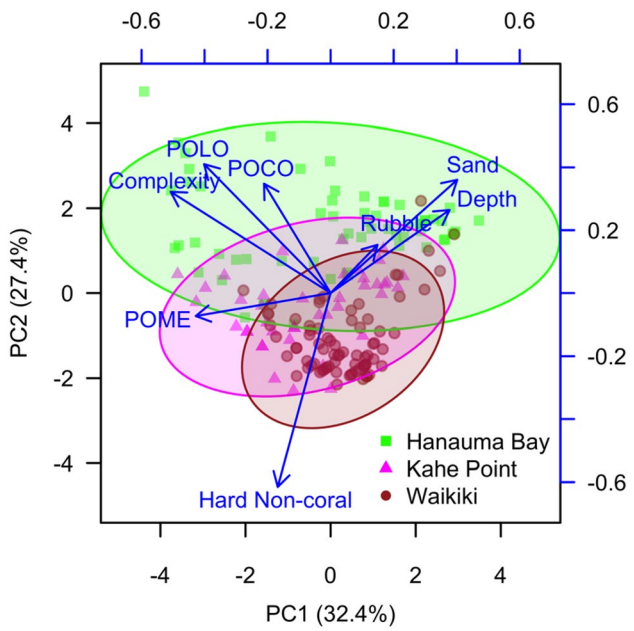

Fig. 2 Principal Component Analysis biplot of the seafloor habitat within 5 m of 180 intact *Pocillopora grandis* coral colonies (261 surveys; Hanauma Bay: 102 surveys, 62 colonies; Kahe Point: 41 surveys, 41 colonies; Waikīkī: 118 surveys, 77 colonies). Black axes are on the scale of the PC scores (surveys), while blue axes reflect the scale of loadings on each axis (habitat variables). Ellipsoids for each site are standard deviations with confidence limits of 0.95. Coral species within 5 m of focal *P. grandis* colonies: POCO = *Porites compressa*; POLO = mounding *Porites* spp.; POME = *Pocillopora meandrina*. Habitat variables are defined in the text and Table S2

(model mean ± SE; 3.65 ± 0.39; Tukey: $z = -2.80$, $p = 0.014$) and invertebrate diversity (1.19 ± 0.08; Tukey: $t = -2.75$, $p = 0.018$), species richness (4.16 ± 0.32; Tukey: $z = -2.79$, $p = 0.014$), and abundance (8.88 ± 0.88; Tukey: $z = -3.41$, $p = 0.002$) per colony compared to colonies at Hanauma Bay (fish abundance: 2.17 ± 0.27; invertebrate diversity: 0.82 ± 0.09; species richness: 2.83 ± 0.26; abundance: 4.87 ± 0.59). Kahe Point colonies also had greater invertebrate abundance (8.46 ± 1.72; Tukey: $z = -2.40$, $p = 0.043$) than those at Hanauma Bay.

Multivariate analysis not including inter-branch space found that few reefscape variables contributed significantly to variation in resident assemblages (Figs. 4, S5) (db-RDA: F = 4.48, $p = 0.001$). Significant predictors included hardbottom (db-RDA: F = 2.75, $p = 0.002$), sand (db-RDA: F = 2.42, $p = 0.003$), and vertical relief (db-RDA: F = 1.67, $p = 0.006$), but each explained less than 1% of the variation according to PERMANOVA. Site (db-RDA: F = 3.07, $p = 0.001$) and depth (db-RDA: F = 4.46, $p = 0.001$), however, each explained 2.5% and 1.8%, respectively, by PERMANOVA.

Colony-scale effects on assemblage structure of intact colonies

Pocillopora grandis colony volume decreased as depth increased (Fig. S6a) (LMM; F = 5.3, $p = 0.02$), albeit mildly ($R^2 = 0.03$), while colony inter-branch spacing increased substantially with depth (Fig. S6c) (LMM; F = 14.7, $p < 0.001$) and colony volume (Fig. S6b) (LMM; F = 62.1, $p << 0.001$). These predictor variables combined explained 32.7% of the variation in inter-branch spacing.

Except for inter-branch spacing, colony-scale variables had comparatively greater effects on assemblage structure of intact *P. grandis* colonies. The effect of colony inter-branch spacing was non-significant in all univariate models, so this predictor was dropped to increase sample sizes. Colony volume (Figs. 3, S3 middle) had significant positive effects on diversity (fishes, LMM: F(1, 200.6) = 124.8, $p << 0.001$; invertebrates, LMM: F(1, 197.0) = 39.2, $p << 0.001$), species richness (Fig. S7b,d) (fish, GLMM: $\chi^2 = 138.2$, $p << 0.001$; invertebrates, GLMM: $\chi^2 = 36.5$, $p << 0.001$), species evenness (fishes, LMM: F(1, 198.6) = 26.1, $p << 0.001$; invertebrates, LMM: F(1, 250.1) = 9.1, $p = 0.003$), and abundance (Fig. S7a,c) (fishes, GLMM: $\chi^2 = 176.4$, $p << 0.001$; invertebrates, GLMM: $\chi^2 = 82.4$, $p << 0.001$). The amount of living colony tissue (Figs. 3, S3 right) was significantly correlated only with metrics of fish assemblages (abundance, GLMM: $\chi^2 = 44.3$, $p << 0.001$; diversity, LMM: F(1, 244.1) = 14.4, $p < 0.001$; species richness, GLMM: $\chi^2 = 23.0$, $p << 0.001$; species evenness, LMM: F(1, 245.1) = 10.0, $p = 0.002$).

To evaluate the effects of colony-scale variables within a multivariate framework, I ran two distance-based redundancy analyses, with and without inter-branch spacing. The model with inter-branch spacing included explained 20.9% of the variation in assemblage structure, albeit across only 2 sites (Fig. S8a,b). Inter-branch spacing had a moderate effect (db-RDA: F = 1.87, $p = 0.022$) compared to living colony tissue (db-RDA: F = 8.02, $p = 0.001$) and volume (db-RDA: F = 2.23, $p = 0.011$), and explained less variation (PERMANOVA: inter-branch, $R^2 = 0.014$; percent alive, $R^2 = 0.067$; volume, $R^2 = 0.018$).

When inter-branch spacing was dropped from the model and colonies from all three sites included, the full model explained 16.7% of the variation (Figs. 4, S5). Colony volume (db-RDA: F = 5.24, $p = 0.001$) and percentage of living tissue (db-RDA: F = 9.93, $p = 0.001$) explained 2.4% and 4.6%, respectively, of total variation in assemblage structure by PERMANOVA, equaling 47.9% of total attributable variation.

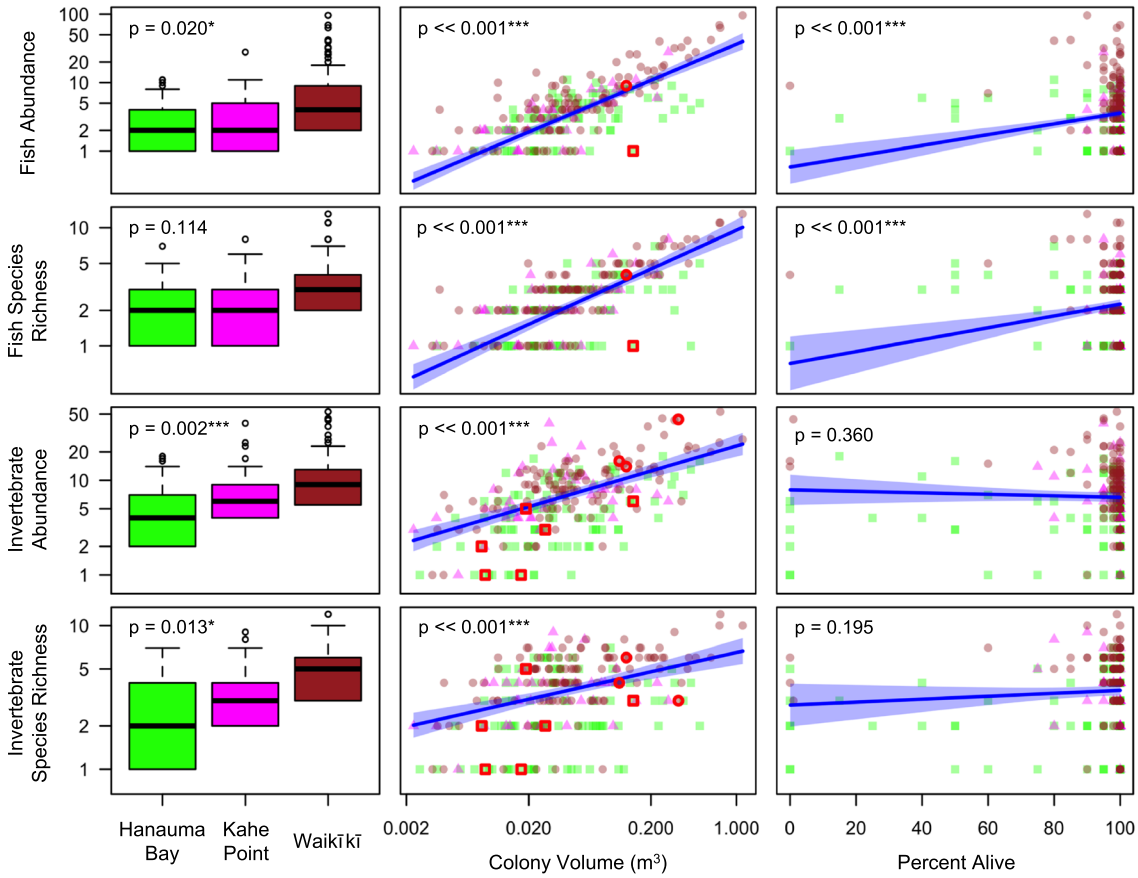

Fig. 3 The statistical effects of site (left), colony volume (middle), and percentage of living coral tissue (right) of *Pocillopora grandis* on resident fish abundance (first row), resident fish species richness (second row), invertebrate abundance (third row), and invertebrate species richness (fourth row) in general linear mixed effects models with colony identity and survey date as random effects. Colony volume and all assemblage metrics (responses) are plotted on log-scale axes. The rows with fish and invertebrate metrics are missing 34 and 8 surveys from the raw data, respectively, due to fish and invertebrate absence and inability to plot zeros on the log scale. See Figure S4 for plots with linear axes that show surveys with fishes and invertebrates absent. Points outlined in red on the invertebrate plots are from surveys of colonies with ≤5% living tissue, indicating that colony life or death did not clearly affect the general patterns

Species co-occurrence on intact colonies

The co-occurrence analysis showed how species associations changed with increasing colony size. All 24 species were present in at least 1 survey of both colony size ranges (Fig. 5; for an analysis with all colonies see Fig. S9). On small colonies, 9% of associations were non-random (Fig. 5a) (7 negative, 18 positive), compared to 15.6% on larger colonies (Fig. 5b) (6 negative, 37 positive).

Of the negative associations, only that between the two hawkfishes (*Paracirrhites arcatus* and *Cirrhitops fasciatus*) remained constant across coral size ranges. Other negative associations between fishes on small colonies included a damselfish (*Plectroglyphidodon johnstonianus*) with another damselfish (*Dascyllus albisella*) and a wrasse (*Pseudocheilinus tetrataenia*), yet both became random on larger corals. Of all positive associations, six were found in both coral size classes, and all involved guard crabs of the genus *Trapezia*. The guard crab *T. intermedia* had only one significant association with another guard crab. By contrast, all other guard crab species were positively associated with 2–5 congeners on larger colonies.

Colony condition and assemblage structure

Due to limitations in the number of surveys of colonies that had experienced some form of disturbance (i.e., detached from sea floor and leaning, split into 2–4 pieces, or highly fragmented), only descriptive analyses were appropriate. Generally, uncommon fishes increased in abundance (Fig. 6a) and richness (Fig. 6b) on the largest colonies and with moderate disturbance, where total fish abundance increased from a maximum of 118 (mean ± SE; 26.3 ± 4.7) on intact colonies to 185 (65.0 ± 23.0) on moderately disturbed colonies. Likewise, fish species richness increased from 13 (6.29 ± 0.59) to 18 (10.1 ± 2.16). However,

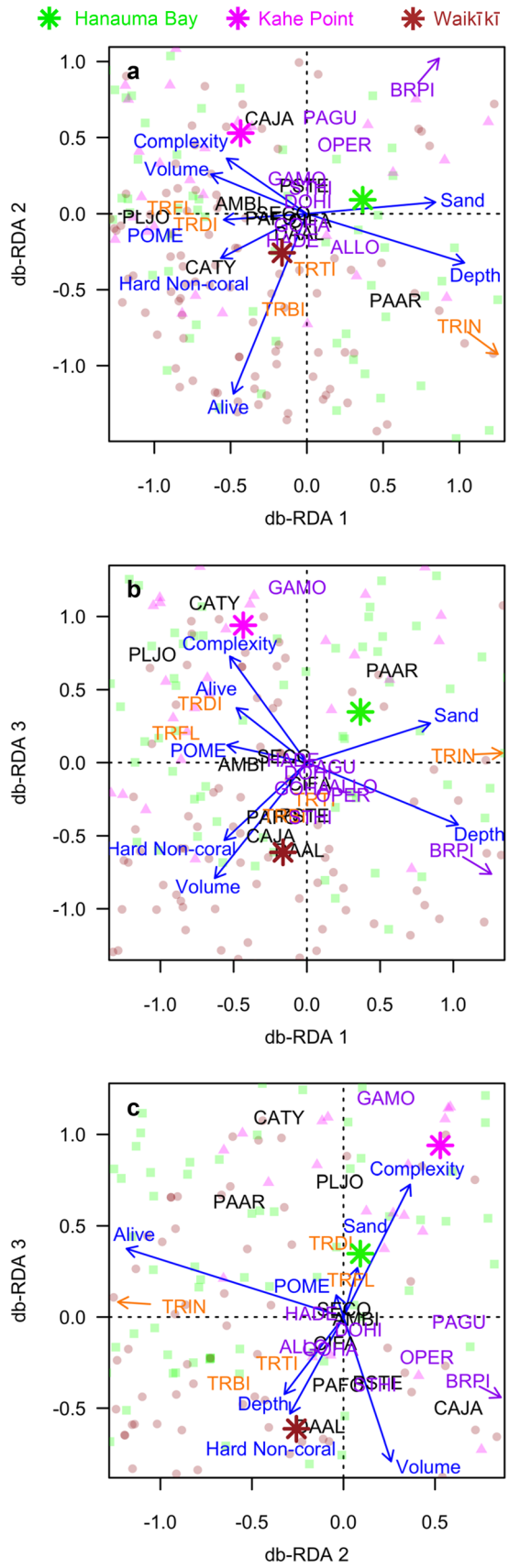

Fig. 4 Distance-based redundancy analysis of 258 surveys of 179 intact *Pocillopora grandis* colony resident assemblages: **a** first and second axes, **b** first and third axes, and **c** second and third axes. Analysis is restricted to species listed in Table 1. The effects of colony variables (Volume, Alive) and environmental variables (Depth, Complexity, Hard Non-coral [hardbottom, dead coral, etc.], Sand, POME = *P. meandrina*) are shown as blue vectors. Values of continuous environmental predictors were scaled by a factor of 1.4 for plotting. Environmental variables were chosen by PCA (see Fig. 2) and used in all models. Centroids of each site are denoted by a large asterisk in its respective color (Hanauma Bay: green; Kahe Point: magenta; Waikīkī: brown). See Table 1 for color and 4-letter species codes. An arrow associated with a species indicates that point occurred outside this zoomed-in plot; see Figure S5 for the full area of each plot

invertebrate abundance (Fig. 6c) (intact: 19.6 ± 2.78; disturbed: 17.9 ± 2.90) and species richness (Fig. 6d) (intact: 5.88 ± 0.49; disturbed: 6.11 ± 0.79) did not drastically increase with moderate disturbance, but like fish assemblages, sharply decreased with high fragmentation. However, local guard crab populations decreased after a lower disturbance threshold. Abundance decreased from a maximum of 22 (9.52 ± 1.04) on intact or leaning colonies to 10 (4.67 ± 1.26) on split or fragmented colonies.

Discussion

Structure of resident assemblages on intact colonies

Prior studies across the tropics have found that the abundance and species richness of resident fishes and invertebrates across a coral reefscape correlate positively with habitat heterogeneity (Wilson et al. 2007) and rugosity (Darling et al. 2017). However, these patterns seem weak if present at all in Hawai‘i. Neither my study of *Pocillopora grandis* nor that of *P. meandrina* by Counsell et al. (2018) found significant support for effects of surrounding reefscape characteristics on resident assemblages on individual colonies. Although these studies explored the assemblage structure of individual coral colonies, the lack of influence by the surrounding reef lends credence to the possibility that habitat in general for fishes and macroinvertebrates may not be limiting on Hawaiian coral reefs, as suggested by Walsh (1987). Nonetheless, Counsell et al. (2018) did find that the abundance and species richness of residents on *P. meandrina* positively correlated with site-scale variables. In this study, site was a significant predictor of local species distributions, fish abundance, and invertebrate abundance, diversity, and species richness, indicating that unmeasured environmental variables such as wave energy (Dollar 1982; Grigg 1983; Franklin et al. 2013) likely contribute to explaining more of the variation in resident assemblages. Depth was also a significant predictor of local species distributions, similar to

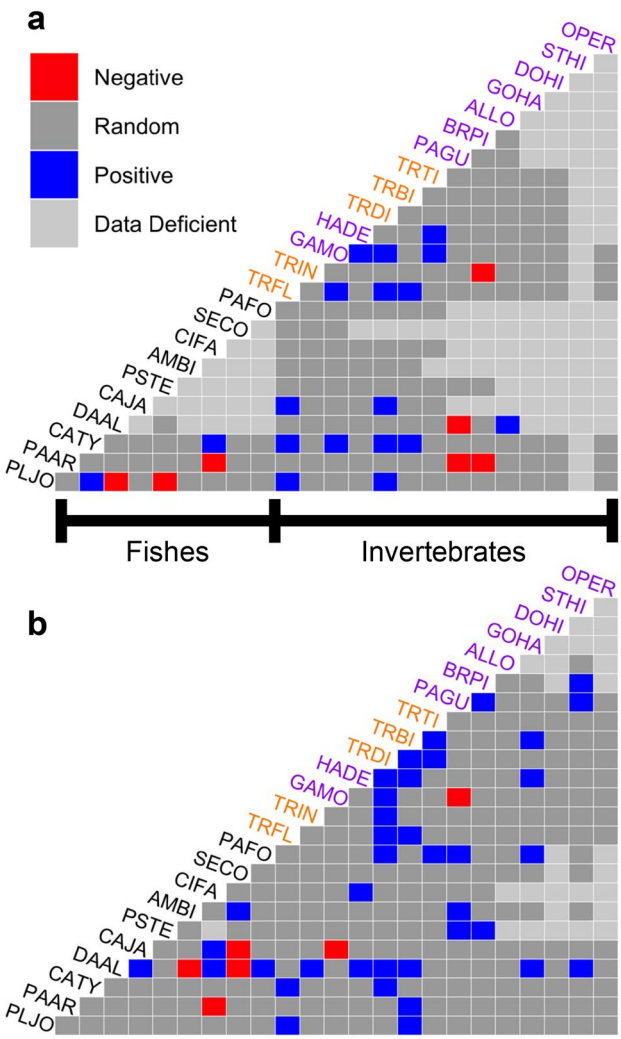

Fig. 5 Probabilistic models of species co-occurrence for **a** small *Pocillopora grandis* colonies (n = 163 surveys, 117 colonies) and **b** large *P. grandis* colonies (n = 95 surveys, 72 colonies; see Table 1 for size categories and species codes). Note that positive associations (i.e., those that happened more often than expected based on random species distributions; blue) and negative associations (less often; red) change as the host coral becomes larger and more habitat space becomes available. See Table 1 for color and 4-letter species codes

previous studies of both invertebrates (Counsell et al. 2018) and fishes (Friedlander and Parrish 1998) in Hawai‘i. The lack of significant effects on coarse metrics of assemblage structure, yet existence of potential species-specific abundance patterns by surrounding reefscape characteristics and depth suggests that some degree of environmental filtering affects *Pocillopora* associated resident assemblages.

Because most species occurred on colonies of all sizes, trends in assemblage structure were likely affected by a combination of limited recruitment and priority effects. First, the extremely limited variation attributable to reefscape variables could be caused by recruitment limitation (Doherty 2002), whereby local population densities are limited by low or patchy larval input rather than density-dependent processes. This scenario has been hypothesized previously for Hawaiian coral reefs (Walsh 1987) such that habitat for reef fishes may not be limiting. Second, priority effects (Almany 2003) and interactions like competition (Bonin et al. 2015) and predation (Hixon 2015) may also be responsible for obscuring any potential effects of surrounding reefscape. The Blue-eye Damselfish (*Plectroglyphidodon johnstonianus*), a territorial interference competitor (Gajdzik et al. 2016), and the Arc-eye Hawkfish (*Paracirrhites arcatus*), a ubiquitous mesopredator of small fishes (Stier et al. 2013) and invertebrates (Hobson 1974) that prefers branching pocilloporids (Kane et al. 2009), were the two most commonly occurring fish species on *P. grandis* across the entire colony size range. This study found evidence for negative interactions between these and other common species on smaller colonies, indicating that these two species may limit colonization by other species until ample space opens via colony growth. Limitations on coexistence likely relax on the largest colonies, which supported a high abundance and species diversity of residents, especially fishes.

Colony inter-branch spacing affected only species-specific distribution patterns and not overall metrics of assemblage structure, similar to the results of Counsell et al. (2018) in Hawai‘i. Additionally, positive and negative associations were evident within and between species typically found at the base of branches (e.g., the scorpionfish *Sebastapistes coniorta*, the hawkfish *Amblycirrhitus bimacula*) and branch tips (e.g., the damselfish *Dascyllus albisella*, the puffer *Canthigaster jactator*). These patterns may be evidence for some unmeasured characteristic of *P. grandis* that affects species habitat preferences, leading to habitat partitioning by colony morphology, similarly found to be a contributing factor for local distributions of two goby species (Pereira and Munday 2016).

Nonetheless, most variation in assemblage structure was explained by colony volume and living tissue, mirroring the findings of Counsell et al. (2018) for assemblages associated with *P. meandrina* around O‘ahu. Colony size has been linked to resident assemblage structure on pocilloporid corals previously (Abele and Patton 1976), and I found that all fish and invertebrate assemblage metrics increased with colony volume. Comparatively, the percentage of living tissue had a much greater effect on fish compared to invertebrate assemblages. Using experimental plots of transplanted *P. grandis* colonies with a gradient of total living coral tissue, Holbrook et al. (2008b) found that fish abundance and species richness, like this study, significantly decreased with live coral. Further, obligate species never colonized completely dead plots. However, contrary to this study, a large decline in fish species richness occurred only with 0% live coral in Holbrook et al. (2008b), providing support for

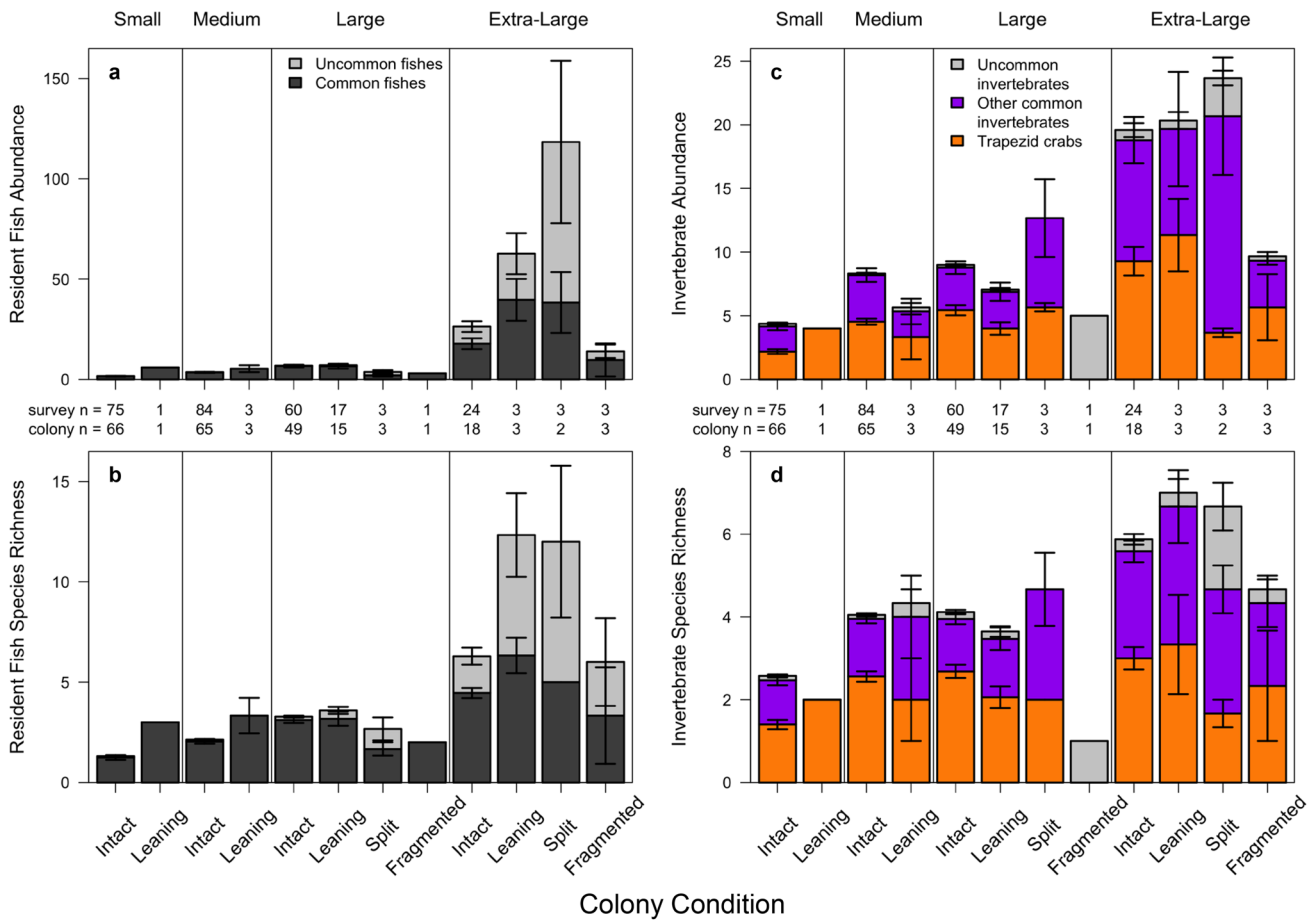

Fig. 6 Bar plots of **a** resident fish abundance, **b** resident fish species richness, **c** invertebrate abundance, and **d** invertebrate species richness by *Pocillopora grandis* colony condition and size. Only surveys of colonies with > 75% living colony tissue are included. No split or fragmented corals were surveyed in the small or medium colony size classes. Bars are means with standard error of each animal group (gray: uncommon fishes/invertebrates that did not occur in at least 5% of surveys; black: common fishes; purple: other common invertebrates; orange: trapezid guard crabs). See Table 1 for a list of all common species and 4-letter species codes

potential scale-dependent effects. While the physical structure of dead colonies can still support fish assemblages, excluding obligate species directly reliant on live coral tissue, this may not be apparent on the scale of a single coral colony.

Any relationship between living tissue and invertebrate assemblage metrics was likely affected by the influx of uncommon species on larger colonies with lower levels of live tissue, shifting assemblage composition but leaving total abundance and species richness unchanged. In a study relating live colony tissue and decapod diversity on *P. meandrina* in Hawai‘i, Coles (1980) found that obligate symbiont decapod diversity was positively correlated with live coral surface area. Dead surface area, however, was associated with greater diversity and number of facultative non-symbiont species. After a crown-of-thorns seastar (COTS) outbreak in Mo‘orea, similar trends were seen in assemblages on *P. grandis*. Reductions in live colony tissue led to greater species alpha-diversity, and large colonies were left as the primary survivors, acting as oases for obligate guard crabs (Leray et al. 2012). Large, live colonies of *P. grandis* may, therefore, act as beacons of desirable habitat for trapezid guard crabs.

Patterns of trapezid guard crab abundance and species richness exhibited similarities and differences to assemblages on *P. meandrina* (Counsell et al. 2018). First, *Trapezia intermedia* exhibited distinct ecological patterns from others, inhabiting smaller and more isolated colonies. This species is likely a pioneer of new habitat that is displaced quickly (Counsell et al. 2018). Second, *T. flavopunctata*, the largest species and least abundant on *P. meandrina*, was the most abundant species on *P. grandis*. Most adult guard crabs occur in single mating pairs on pocilloporid colonies (Preston 1973), but I regularly found multiple pairs inhabiting the same large *P. grandis* colony. Third, species depth associations were similar to that found on *P. meandrina*,

namely *T. digitalis* in the shallows and *T. tigrina* on deeper corals (Counsell et al. 2018). Further, *T. flavopunctata*, a large species typically found at the branch tips, seemed to associate more with *T. digitalis*, a smaller species found near the base. A similar pairing between *T. tigrina* and *T. bidentata* suggests intra-colony microhabitat partitioning. Across all 502 surveys, only 7 found every species coexisting, and only 2 extraordinarily large colonies contained adults from each. Lastly, although observed on colonies with only 5% living tissue in this study, guard crab abundance and species richness declined with live colony tissue and size (Coles 1980; Counsell et al. 2018). Except on the largest colonies, facultative species replaced guard crabs once live tissue declined to 25%.

Understanding what shapes patterns of guard crab abundance and species richness is critically important due to host coral – resident mutualisms. Guard crabs alleviate the effects of sedimentation (Stier et al. 2012) and defend their host coral from invertebrate corallivores (McKeon et al. 2012). Different species also better defend against different predators due to differences in size, so greater crab species richness confers greater protection (McKeon and Moore 2014).

Structure of resident assemblages on broken colonies

Exposed Hawaiian reefs are strongly affected by seasonal large waves (Grigg 1983) that play a role in structuring reef fish assemblages (Friedlander and Parrish 1998). Coral species differ in mechanical strength, so wave strength can greatly affect coral community structure at larger spatial scales (Storlazzi et al. 2005). Rodgers et al. (2003) showed that *Pocillopora meandrina* has a high skeletal strength, which greatly reflects the environment where it typically occurs in Hawai‘i (Franklin et al. 2013). *P. grandis* may have even greater skeletal strength (pers. obs.), but large colonies are supported by a narrow base, so colony dislodgement and fragmentation by waves increase with colony size. Barring unusually large disturbance events, this species grows until it can no longer support its own weight and collapses, but like *P. meandrina* (Rodgers et al. 2003), these colonies can remain alive in a broken state for extended periods of time (pers. obs.). Only highly fragmented colonies suffered rapid mortality, likely due to predation by cushion stars commonly found along the south shore of O‘ahu (Glynn and Krupp 1986). Rodgers et al. (2003) found low mortality of small *P. meandrina* fragments in Kāne‘ohe Bay, Hawai‘i, but cushion stars there are very rare (Glynn and Krupp 1986).

Effects of physical disturbance on resident fish and invertebrate assemblages were most notable on the largest colonies, as these were more likely to experience fragmentation. Colonies exposed to moderate disturbance (i.e., detachment from seafloor but remaining at least relatively intact) acquired expanded fish assemblages that included surgeonfish, squirrelfish, and cardinalfish species not common on intact colonies. In an experimental manipulation of coral disturbance, Syms and Jones (2000) found an influx of uncommon species on the largest disturbed colonies, reflecting the importance of habitat heterogeneity and creation of new microhabitat types that attract more rubble-associated species. However, both fish and invertebrate abundance and species richness drastically declined with full colony fragmentation, matching previous studies that showed more extreme levels of habitat disturbance cause the greatest changes in fish assemblages (Syms and Jones 2000; Yahya et al. 2011).

Potential for coral restoration

Pocillopora grandis occupies a unique role on modern Hawaiian reefs. As part of a healthy reef (e.g., Hanauma Bay) with later successional stage coral communities, they contribute substantially less to the overall structural complexity of the reefscape, attracting only typical resident species. With potentially lower levels of recruitment, resident fish and invertebrate abundance could be diluted by greater habitat availability. However, on degraded reefs (e.g., Waikīkī), large colonies contribute a relatively large proportion of the structural complexity, creating oases of life in an otherwise flat and relatively barren reefscape. Hawaiian reefs are naturally depauperate of coral species relative to other Indo-Pacific regions (Grigg 1983), and *P. grandis* supports greater reef fish diversity than the vast majority of other Hawaiian coral species (Coker et al. 2014).

The capability of larger colonies to support concentrated assemblages with mutualist species on degraded reefs makes *P. grandis* ideal for coral restoration. Naturally occurring fish mutualists can increase coral growth rates (Holbrook et al. 2008a, 2011) and resilience to thermal stress (Shantz et al. 2023). Therefore, outplanted *P. grandis* colonies could support greater assemblages soon after transplantation that enable colonies to better survive thermal stress events, contributing to greater long-term restoration success. Indeed, fragmented branches of *P. grandis* outplanted to artificial structures off Waikīkī have grown substantially faster than transplanted fragments and naturally occurring colonies of other species (pers. obs.).

Additionally, the ability of *P. grandis* to become detached from the seafloor yet live for years and host a greater variety of species makes these corals uniquely suited to the higher average wave energy associated with Hawaiian reefs. Highly fragmented colonies also provide natural sources of fragments ideal for outplanting, especially given (1) their large and morphologically complex branches, (2) ability to keep or quickly acquire guard crabs, and (3) high survival and growth post-transplantation (pers. obs.). Given the vast array of threats facing coral reefs, it is imperative that coral

restoration efforts use species that are adaptable for changing conditions and able to host abundant mutualist fishes and invertebrates, both of which should enhance reef resilience.

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Declarations

Conflict of interest The author has no relevant financial or non-financial interests to disclose.

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